ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

(i) f_x = 16/0.005 = 3200 pixels. (ii) Sensor width = 6.4 mm. I used W = 2Z tan(f/(2Z)) and obtained about 16 mm; pixel scale = 16/1280 = 0.0125 mm/pixel.

Q1(b) [3 marks]

The smallest resolvable force is 0.122 N.

Q1(c) [3 marks]

Using forward Euler, F approximately 1 + AT = 0.8 and G approximately BT = 0.1. Hence x[k+1] approximately 0.8x[k] + 0.1u[k].

Question 2 [10 marks]

Q2(a) [2 marks]

At equilibrium the velocity should be zero. The torque and gravity torque must balance.

Q2(b) [2 marks]

$$A = [[0, 1], [-61.3125, -2.5]], B = [[0], [50]]$$

Q2(c) [4 marks]

The balance is 50(90-θss) = 2.4525 sin θss. Solving gives a small error of about 0.05°.

Q2(d) [2 marks]

Gravity compensation makes the steady-state error zero. It does not change the closed-loop poles because it only adds a constant torque.

Question 4 [10 marks]

Q4(a) [2 marks]

Closed-loop characteristic polynomial is $s^2+(a+bk_2)s+(c+bk_1)=s^2+4s+8$, giving poles -2±2j. At DC, $\theta/r = bN_{ff}/(c+bk_1)$, so unity gain requires $N_{ff}=(c+bk_1)/b=(6+2)/1=8$.

Comparing coefficients with $s^2 + 4s + 8$ gives $a + bk_2 = 4$ and $c + bk_1 = 8$. The feedforward gain is chosen from the nominal plant so the reference scaling is unity; here this gives $N_{ff} = 8$. The closed-loop roots of $s^2 + 4s + 8$ are $-2 \pm 2j$.

Q4(b) [2 marks]

With actual b=0.75 and unchanged $N_{ff}=8$, $k_1=2$, the steady-state relation is $\theta_{ss} = bN_{ff}/(c+bk_1) = 0.75x8/(6+0.75x2)=6/7.5=0.8$. For r=1, percentage error = (1-0.8)x100%=20%.

With the actual gain, the steady-state output is $y_{ss} = b_{actual}N_{ff}r/(c+b_{actual}k_1)$ for the stated setup. Substituting the given numbers gives $y_{ss} = 0.8$ for a unit reference. Therefore $e_{ss} = 1 - 0.8 = 0.2$, i.e. a 20% error.

Q4(c) [4 marks]

The characteristic polynomial is $s^3+(a+bk_2)s^2+(c+bk_1)s+bK_I$. I use the target $s^3+6s^2+12s+16$, giving $k_2=4$, $k_1=6$, $K_I=16$.

Q4(d) [2 marks]